Closed-Loop Bayesian Optimization of Multi-Material 3D-Printed Tensegrity-Inspired Energy Absorbers

Extended Abstract — ASME IDETC-CIE 2026
Track DAC-10: Design of Engineering Materials and Structures

Abstract

Tensegrity-inspired architectures—rigid struts suspended within a continuous flexible network—can exhibit tunable nonlinear force–displacement responses and favorable energy absorption per unit mass, motivating their use in lightweight energy-absorbing and protective structures [1, 9]. Multi-material fused deposition modeling (FDM) can co-print rigid (PLA) struts and flexible (TPU) elements in a single build [4, 10], but the resulting design space—strut geometry, tension-element cross-section, connectivity topology, and unit-cell tiling—is too large to explore by trial and error. We present a closed-loop experimental campaign that uses Bayesian optimization (BO) [8] to drive a design–print–test workflow operating directly on physical measurements, without relying on calibrated finite-element simulation ~~[8]~~for objective evaluation.

1 Introduction and Motivation

Tensegrity structures combine compression-only members with a pretensioned tension network to achieve stiffness without rigid joinery [9]. Their nonlinear, often load-limiting force–displacement response makes them attractive for impact mitigation, packaging, and wearable protective gear [1]. Idealized cable–strut tensegrities are difficult to manufacture at scale, but *tensegrity-inspired* architectures fabricated by multi-material FDM ~~retain much of the desirable mechanical behavior, reproducing load-limiting and progressive-collapse responses while~~ can reproduce key tensegrity behaviors—including post-buckling stability and load-limiting force–displacement response—while enabling monolithic fabrication ~~[4, 7]~~[7]. PLA provides rigid struts; TPU provides flexible, rate-dependent tension elements whose ~~viscoelastic response can increase~~ viscoelasticity is expected to contribute hysteretic energy dissipation under impact loading, consistent with the energy-absorbing behavior reported for rigid–TPU multi-material prints [4]. The challenge is that small geometric and topological changes can produce large changes in the measured response, and high-fidelity simulation of multi-material FDM specimens—including interfacial slip, print defects, and TPU rate-dependence—is expensive and often unreliable [6]. Pure trial-and-error sweeps over the joint design space are also infeasible: even modest parameter resolutions yield thousands of candidates, each requiring print and test time. We instead treat the physical experiment as the source of truth and use BO to select which specimens are most informative to fabricate next—an approach used in related closed-loop materials and structures campaigns [5, 6]. Unlike prior tensegrity-inspired studies that evaluate a fixed design set [4, 7], we adaptively select designs using parallel, noisy multi-objective BO driven by measured F_{peak}, SEA, and η.

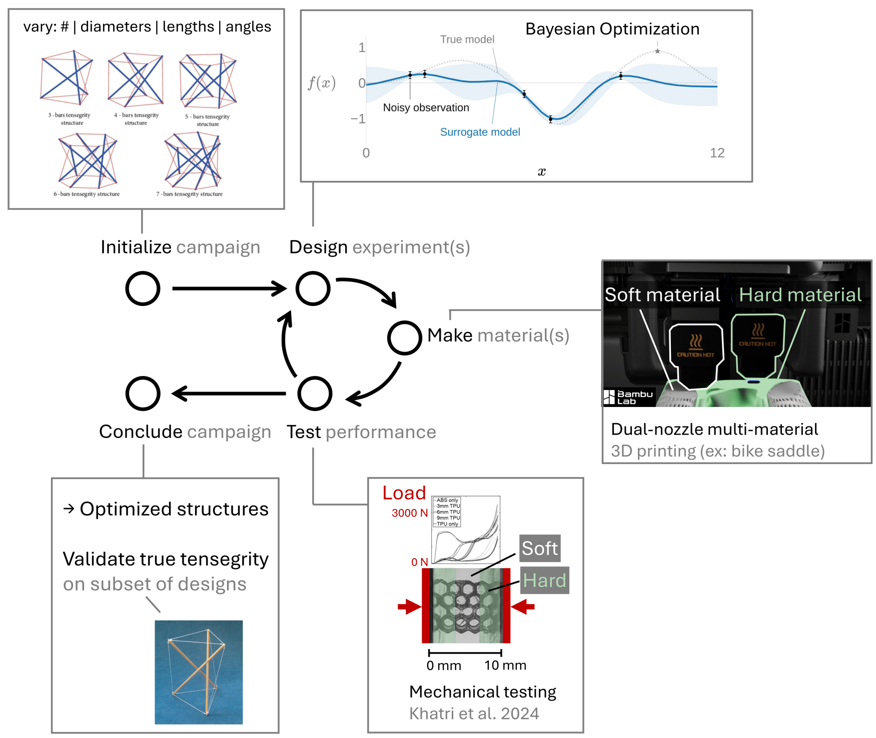

Figure 1: Closed-loop, BO-driven design–print–test workflow. Candidate tensegrity-inspired unit cells are instantiated from a parameter vector, fabricated by multi-material FDM, and tested under quasi-static compression and drop-weight impact. Measured peak force, specific energy absorption (SEA), and compaction efficiency update a Gaussian-process surrogate that proposes the next batch of designs.

2 Approach

2.1 Closed-Loop Workflow

Figure 1 summarizes the campaign. Each iteration proceeds in four steps: (i) a parameterized unit-cell design is instantiated from a candidate vector $\mathbf{x}$; (ii) the specimen is sliced and printed on a multi-material FDM system; (iii) it is tested under quasi-static compression and instrumented drop-weight impact; and (iv) the resulting performance metrics update a Gaussian-process (GP) surrogate, which a batch acquisition function queries to recommend the next set of candidates. Surrogate updating and experiment selection are automated; specimen handling, slicing, and test setup remain manual.

2.2 Design Space

We parameterize a family of unit cells using a core-wrapping architecture in which rigid PLA struts are enclosed by continuous TPU skins~~to improve interfacial interlocking and reduce delamination~~; the wrapping rationale follows multi-material FDM strategies in which rigid panels are wrapped by stretchable soft material to improve load transfer at the rigid–flexible interface [10]. The design vector $\mathbf{x}$ includes strut diameter and length, tension-element width and thickness, strut count per unit cell, discrete connectivity topology, and unit-cell tiling pattern~~(mixed continuous–discrete, $d \approx 8$ variables). Bounds~~, giving a mixed continuous–categorical space of roughly eight variables. Categorical choices (topology, tiling) are encoded explicitly rather than embedded in a continuous metric, and bounds on the continuous variables are chosen to remain within the printer's resolution and overhang limits and to keep specimens in a common form factor for the test fixtures.

2.3 Surrogate and Acquisition

The GP surrogate is trained on three measured performance metrics: peak transmitted force F_{peak}, specific energy absorption (SEA), and compaction efficiency η. We use independent GPs per objective with Matérn kernels; hyperparameters are fit by marginal likelihood with priors to avoid pathological length scales at small n. ~~Categorical design choices (topology, tiling) are encoded explicitly so that the kernel respects the discrete structure of the design space.~~ We use q-noisy expected hypervolume improvement (qNEHVI), which explicitly accounts for observation noise in parallel multi-objective BO [3], to select the next set of candidates. Print failures are ~~modeled~~ treated as a probabilistic feasibility constraint ~~and~~ incorporated into constrained acquisition~~to limit sampling in infeasible regions~~, following recent constrained multi-objective practice in self-driving labs [5]. ~~To mitigate numerical pathologies in~~ Where applicable, we use numerically stable log-space variants of improvement-based acquisitions to mitigate vanishing-gradient pathologies as data and constraints accumulate~~, we apply numerically stable log-space reformulations~~ [2].

3 Methods

3.1 Fabrication

Specimens are fabricated on a multi-material FDM system capable of co-printing PLA and TPU within a single build [10]. Print parameters (temperature, line width, infill, retraction) are held fixed within a batch to isolate the effect of the design variables. Each specimen is weighed and dimensionally inspected prior to testing so that mass and geometric deviations are available as covariates.

3.2 Mechanical Testing

Quasi-static compression is performed on a screw-driven load frame at a fixed nominal strain rate, with force–displacement recorded to densification. Drop-weight impact testing uses an instrumented tup at a fixed impact energy; transmitted force is captured at high sample rate, and specimens are imaged before and after impact to document failure modes. From each test we extract F_{peak}, SEA per unit mass, and compaction efficiency η, defined consistently across the campaign. A control specimen from an earlier batch is retested periodically to monitor ~~and correct for~~ TPU batch-to-batch and printer drift.

4 Expected Outcomes and Discussion

We will report how BO-selected designs evolve across iterations, how GP surrogate predictions and uncertainties change as data accumulate, and which Pareto-efficient trade-offs emerge between SEA and peak transmitted force within the explored design space. We will also discuss practical lessons for operating the closed-loop workflow, including handling of print failures, batch-to-batch variability in TPU response, the influence of the rigid–flexible interface on energy absorption, and the exploration–exploitation balance of the acquisition function under realistic experimental noise. We expect the campaign to clarify where physical experimentation is genuinely required versus where simpler analytical or multi-fidelity surrogate shortcuts [6] could be substituted in future work.

5 Conclusion

This work demonstrates a ~~largely~~ partially automated, experiment-driven closed-loop pipeline for designing multi-material 3D-printed tensegrity-inspired energy absorbers, with surrogate updating and candidate selection automated and fabrication and testing kept manual. By updating a GP surrogate directly from physical measurements, the workflow ~~sidesteps the calibration burden of high-fidelity simulation~~ avoids direct dependence on calibrated finite-element simulation for objective evaluation while still providing principled, multi-objective design recommendations. The ~~methodology is applicable~~ same closed-loop pattern is expected to transfer to other multi-material additively manufactured architectures whose performance is dominated by hard-to-simulate effects.

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